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CALIBRATION PROCEDURE

Standard Masses, Class M2 — per OIML R111-1 and EURAMET cg-18 Mass Comparison Principles

1. Scope & Application

This procedure applies to the calibration of weights and standard masses of accuracy class M2 (and lower classes M1, M2-3, M3 by extension), in nominal values from 1 mg to 50 kg, per OIML International Recommendation R111-1, "Weights of classes E1, E2, F1, F2, M1, M1-2, M2, M2-3 and M3". The mass comparison principle follows the general approach described in EURAMET calibration guidance for mass standards.

2. Reference Standards & Traceability

STANDARD	DESCRIPTION	TRACEABILITY
Class F1 Reference Weight Set	At least one accuracy class better than the M2 item under test, covering the required nominal range	National metrology institute, ≤ 36 months
Mass Comparator	Resolution and repeatability appropriate to the nominal value and class M2 tolerance	Internal verification against reference standards
Reference Thermo-Hygrometer & Barometer	Ambient conditions for air density / buoyancy correction	Accredited lab, ≤ 12 months

3. Environmental Conditions

PARAMETER	REQUIREMENT
Temperature	Stable laboratory environment; weights and comparator allowed to reach thermal equilibrium before comparison (minimum 2 hours for items moved from storage)
Handling	Weights handled only with gloves, forceps, or lifting tools appropriate to class — never bare hands
Air Currents / Vibration	Comparator located on a stable bench, free from draughts and vibration

4. Procedure

- Visual Inspection.** Inspect each weight for corrosion, damage, surface contamination, or loss of adjustment cavity sealing. Reject from calibration any weight with damage likely to affect mass stability; record findings.
- Cleaning.** Clean weights per the method appropriate to their class and material, avoiding any process that could alter surface mass.
- Mass Comparison.** Compare the weight under test against the reference standard(s) of equal or adjacent nominal value using an ABA or ABBA substitution weighing sequence on the mass comparator, recording the comparator indication at each substitution.
- Repeat Determinations.** Repeat the comparison sequence a minimum of 3–5 times to establish repeatability (standard deviation) of the comparison process at this nominal value.

5. **Air Buoyancy Correction.** Record ambient temperature, pressure, and relative humidity at the time of comparison and calculate the air density. Apply a buoyancy correction to the comparison result using the conventional mass density of the test weight (8000 kg/m³ unless the actual material density is known and stated).
6. **Conventional Mass Calculation.** Calculate the conventional mass of the weight under test as the reference value plus the mean corrected difference observed during comparison.

5. Uncertainty & Calculation

The combined standard uncertainty is calculated from: the calibration uncertainty of the reference standard(s); the repeatability (standard deviation) of the comparison process; the resolution of the comparator; the uncertainty of the air buoyancy correction; and any drift in the reference standard since its last calibration. These components are combined per the GUM method to determine the expanded uncertainty (k=2) of the conventional mass value.

6. Acceptance Criteria

The weight passes calibration where the deviation of its conventional mass from its nominal value, together with the expanded uncertainty, falls within the maximum permissible error (MPE) for class M2 as tabulated in OIML R111-1 for the relevant nominal value. Weights exceeding the class M2 MPE are reported as out of tolerance and may be downgraded to a lower class or withdrawn from service.

Prepared By

Technical Manager

Reviewed By

Quality Manager

Approved By

Managing Director